

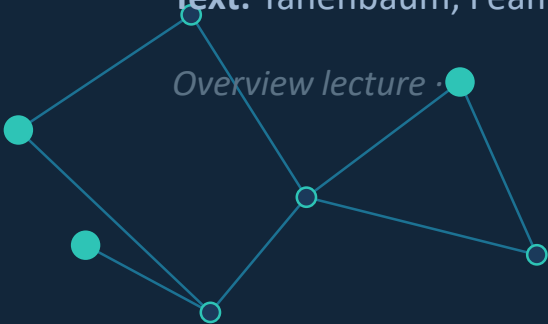
CHAPTER 1

Computer Networks

An Introduction & Course Roadmap

Text: Tanenbaum, Feamster & Wetherall — Computer Networks, 6th ed.

Overview lecture



How the course runs

20%

Quizzes
of final grade

35%

Mid-semester
of final grade

45%

End-semester
of final grade

Textbook & materials

- Tanenbaum, Feamster & Wetherall, Computer Networks, 6th ed.
- Slides and notes posted on Google Classroom
- Join code shared in class today

Assessment is continuous — quizzes reward keeping up week to week, not cramming.

What we'll cover

1 Uses of networks

2 Network hardware

3 Network software

4 Reference models

5 Example networks

6 Standardization

7 Metric units

Who uses networks — and how it looks in 2026

B Business

- Shift from local client–server to cloud & data centers
- SaaS + collaboration suites (docs, chat, video)
- E-commerce and machine-to-machine APIs

H Home & personal

- Streaming video/audio dominates home traffic
- Social platforms, messaging, cloud gaming
- Smart-home & IoT devices always connected

M Mobile

- 5G & Wi-Fi 6E/7 for high-speed wireless
- App-centric: video, maps, payments, wearables
- Sensors everywhere — GPS, health, location

Classic model: sharing resources over the network. The resources just moved — from the server down the hall to the cloud and the content-delivery edge.

Wireless and mobile are related — but different

Wireless	Mobile	Typical 2026 example
No	No	Desktop / workstation wired to office LAN
No	Yes	Laptop docked by Ethernet in different rooms
Yes	No	Wi-Fi 6E in a home or unwired building
Yes	Yes	Phone / wearable / connected vehicle on 5G

The distinction that matters

Wireless = no cable. Mobile = the device moves between networks. A device can be one, both, or neither — and each combination raises different engineering problems (signal vs. handover / addressing).

Social, legal & policy issues

1 Net neutrality

- Should ISPs treat all traffic equally?
- A live, shifting question — not settled
- U.S. federal rules struck down (2025); now a state-by-state patchwork

2 Privacy & data

- Web tracking, profiling, targeted ads
- Regulation: GDPR and similar regimes
- Data ownership & cross-border transfer

3 Security & harm

- Ransomware, DDoS at scale, phishing
- IoT & supply-chain vulnerabilities
- Botnets and large-scale fraud

4 Content & speech

- Copyright & takedowns (DMCA)
- Censorship, moderation, anonymity
- AI-generated content & provenance

Note: net neutrality's legal status has flipped repeatedly (2015 → 2017 → 2024 → 2025). We treat it as an open policy debate, not a fixed rule.

Networks classified by scale

Scale	Type	Typical technology (2026)
Body / desk	PAN	Bluetooth, NFC, wearables
Building	LAN	Wi-Fi 6E / 7, switched Ethernet
City	MAN	Fiber (FTTH/PON), cable, 5G fixed wireless
Country	WAN	ISP backbones, MPLS, SD-WAN overlays
Planet	Internet	The network of all networks

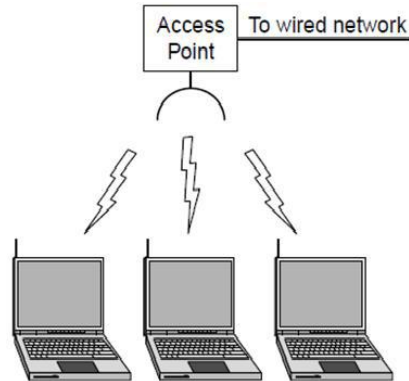
Bigger scale = harder problems: addressing, routing, and who owns which piece. The Internet stitches all of these together.

Local Area Networks

Connect devices within a home or building. In a company, this is the enterprise network.

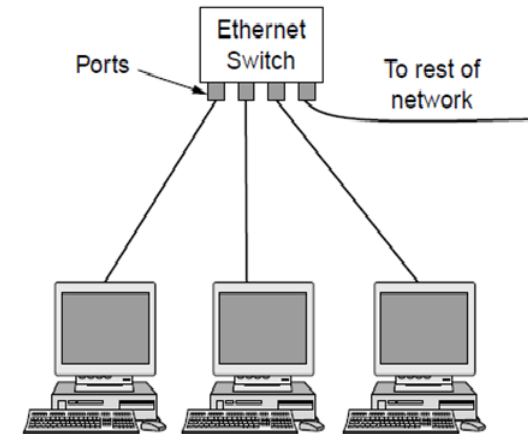
Wireless LAN

- IEEE 802.11 family — Wi-Fi
- Current gen: Wi-Fi 6 / 6E / 7 (802.11ax / be)
- Devices associate with an access point
- 6 GHz band and wider channels boost capacity
- Convenient, but shared airtime & interference



Wired LAN

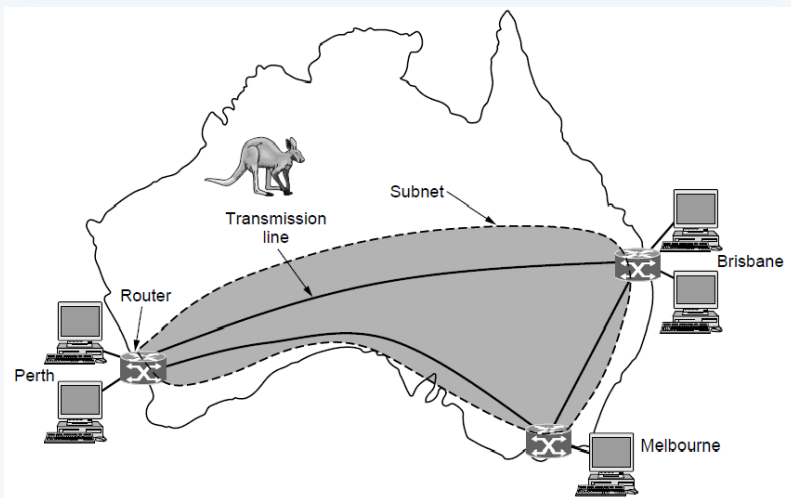
- Switched Ethernet (IEEE 802.3)
- Each device on its own switch port
- Multi-gigabit links now common
- Predictable latency, high reliability
- Backbone for wireless APs and servers



Wide Area Networks, ISPs & overlays

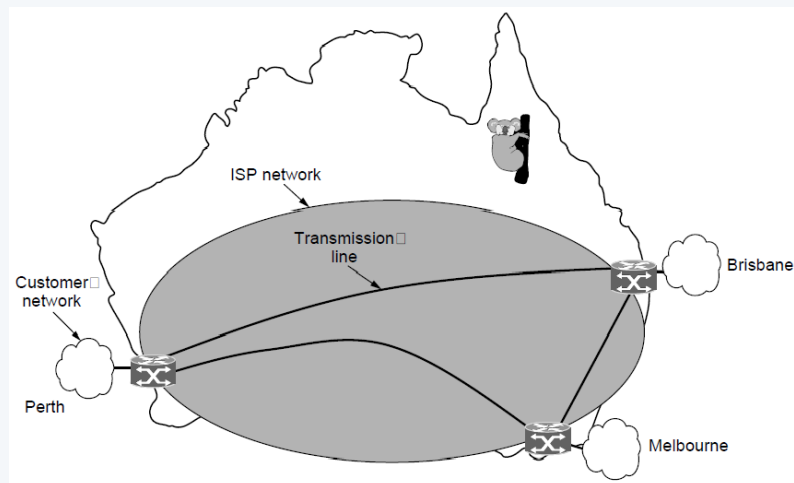
W WAN

- Connects sites across a country/region
- Routing = which path to use
- Forwarding = where a router sends next



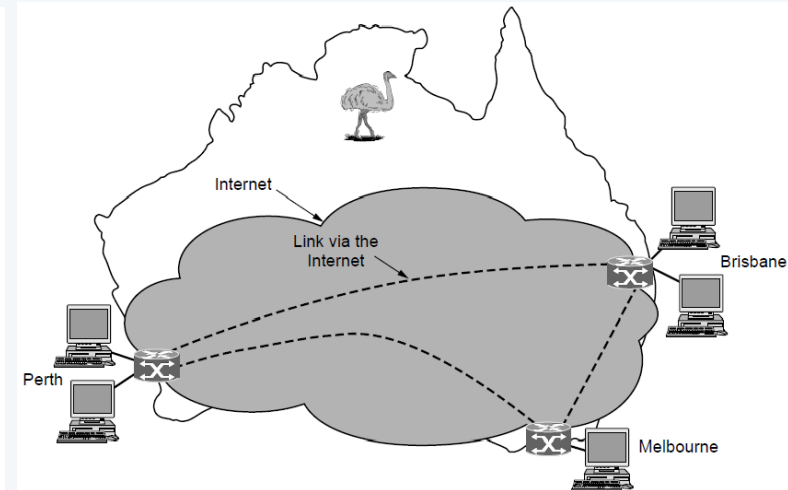
I ISP networks

- Themselves large WANs
- Customers buy connectivity
- Peering & transit link ISPs together



V VPNs & virtual links

- Virtual links running over the Internet
- Reuse shared capacity flexibly
- Basis for secure remote access



Modern twist: software-defined WANs

SD-WAN and network virtualization let operators build and reconfigure wide-area connectivity in software over shared physical links — the same idea as a VPN, generalized. We'll return to routing, forwarding, and virtualization in the network-layer unit.

Protocols & layering

Protocol = an agreement between communicating parties on how to proceed.

Layering = the main way we divide network functions into manageable pieces.

1

Peers Each layer talks virtually to its peer on the other host

2

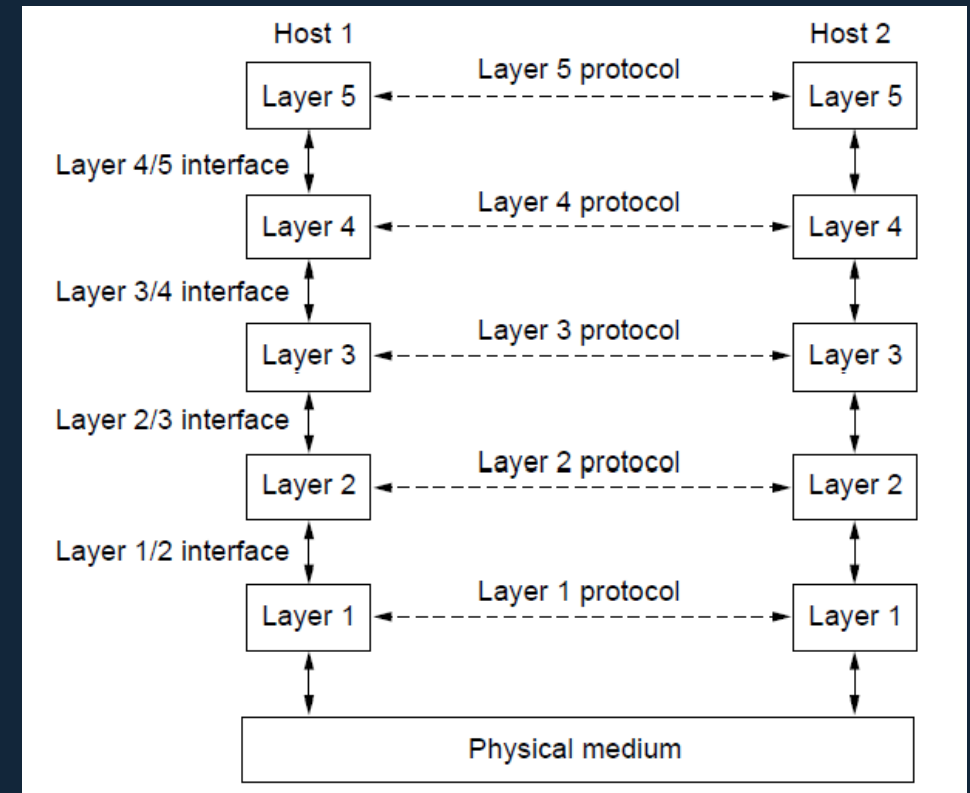
Interfaces A layer uses only the service of the layer below it

3

Services Lower-layer services are reached through an interface

4

Medium At the bottom, bits actually travel over the medium



A metaphor: philosopher – translator – secretary

Each layer serves a different purpose, and can change independently as long as its interface stays the same.

3

Philosophers

Exchange the idea ("I like rabbits") — they don't care what language carries it.

2

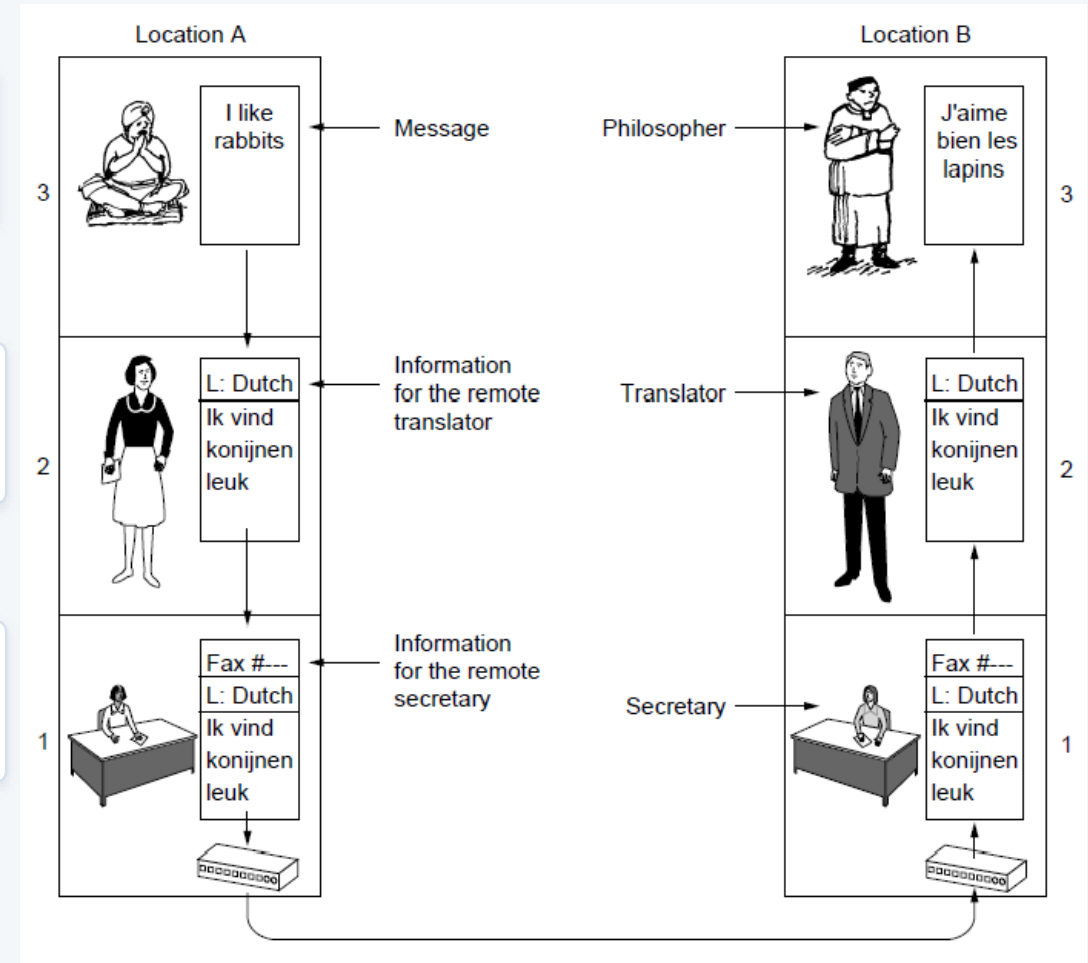
Translators

Agree on a language. They can switch Dutch → Finnish, if both agree and the interface holds.

1

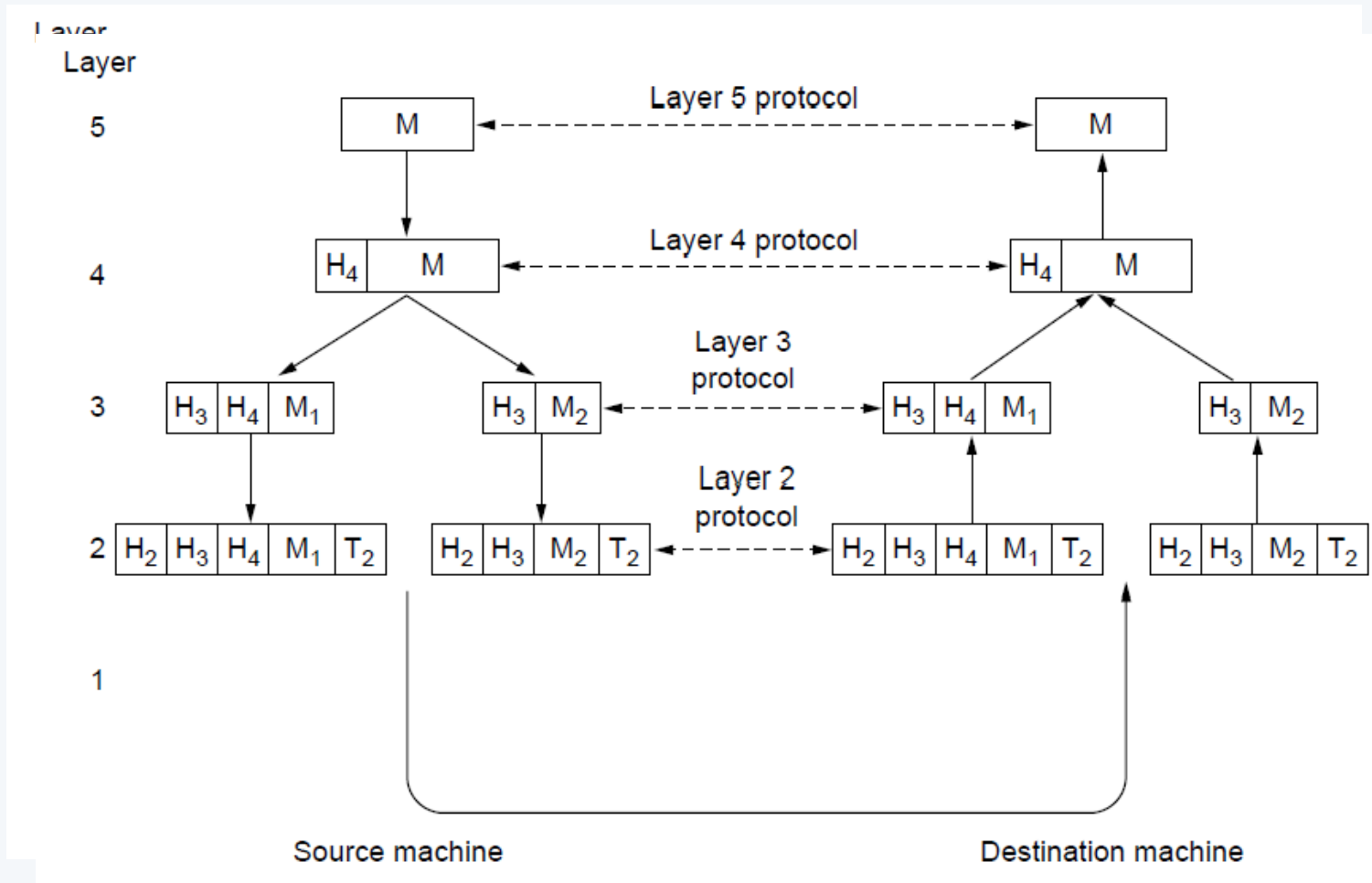
Secretaries

Handle delivery. They can swap fax → email → phone without the layers above noticing.



Encapsulation: each layer adds a header

Going down the stack, each layer wraps the data with its own control info — and strips it on the way up.



Headers (H) carry addresses, sequence numbers, checksums; some layers add trailers (T). Layers may also split or join messages. This is why a captured packet looks like nested envelopes.

Connection-Oriented vs. Connectionless

A service provided by a layer may be one of two kinds:

Service	Example
Connection-oriented — must be set up, then torn down (e.g., phone call)	
Reliable message stream	Sequence of pages
Reliable byte stream	Movie download
Unreliable connection	Voice over IP
Connectionless — messages are handled separately (e.g., postal delivery)	
Unreliable datagram	Electronic junk mail
Acknowledged datagram	Text messaging
Request-reply	Database query



Key terms

Message stream: Message boundaries are preserved

Byte stream: Message boundaries are not preserved

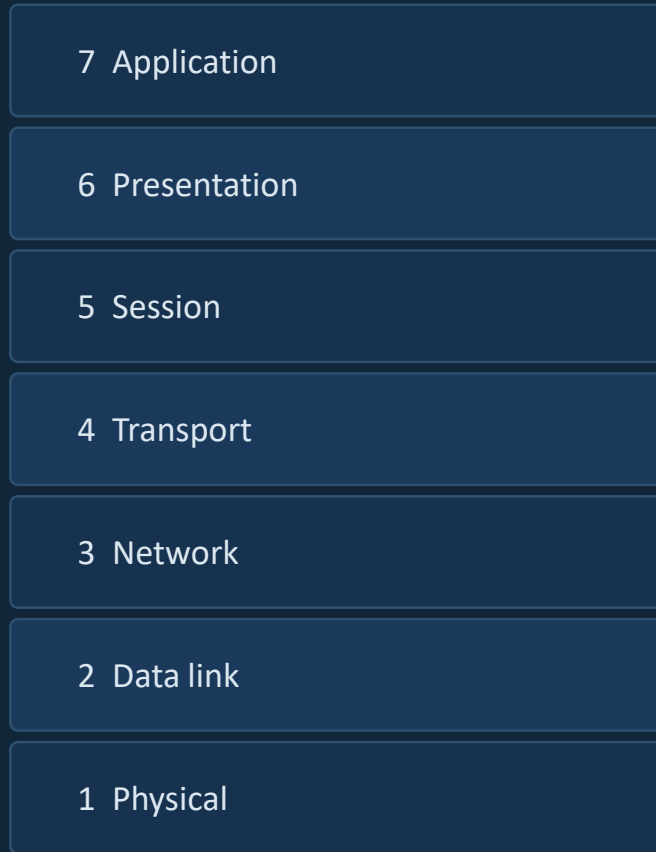
Reliable service: Data is not lost

Acknowledged datagram: Reliable, but connectionless

Request-reply: The sender transmits a single datagram containing a request; the reply contains the answer.

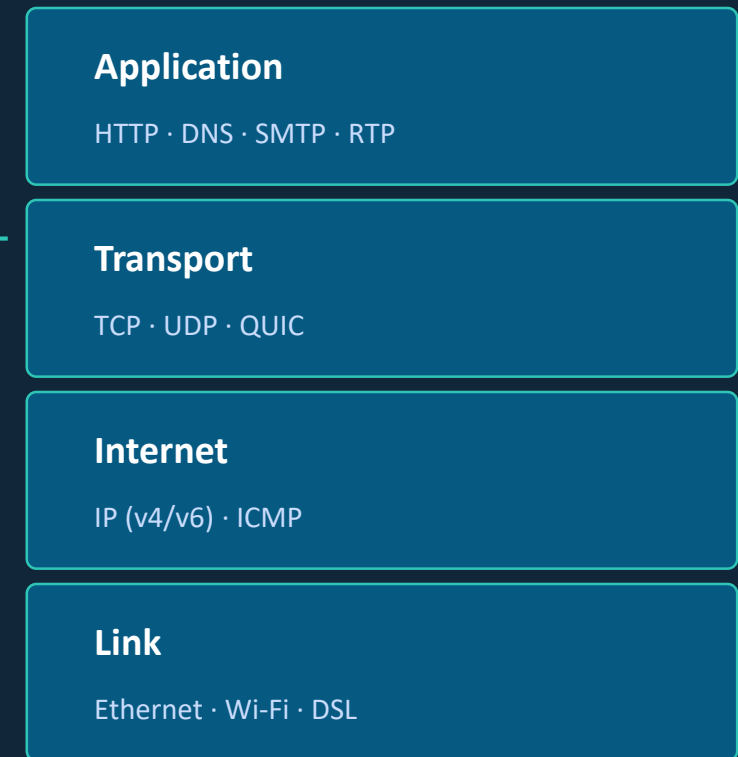
Two reference models

OSI — 7 layers



*Derived from real protocols;
omits OSI's session &
presentation layers.*

TCP/IP — 4 layers



The model we'll use in this course

Based on TCP/IP, but we call out the physical layer explicitly and look beyond just Internet protocols.

5	Application	Apps & their protocols — the Web, email, streaming
4	Transport	End-to-end delivery between processes (TCP/UDP/QUIC)
3	Network	Moves packets across many links (IP, routing)
2	Link	Frames over a single hop (Ethernet, Wi-Fi)
1	Physical	Bits as signals on the medium

Where the clean model bends in practice

The textbook stack is a map, not the territory. Recurring themes this term where reality complicates the layers:

QUIC & HTTP/3

A transport built over UDP at user level —
now much of web traffic

Encryption by default

Nearly all traffic is HTTPS/TLS 1.3; security
isn't an add-on

IPv4 + IPv6

The "narrow waist" is really two coexisting
versions

NAT & middleboxes

Break the classic end-to-end assumption

CDNs & edge

Most "servers" you reach are a nearby
cache

Cloud & data centers

Huge internal networks behind every app

Networks we'll study up close

1 The Internet

- A network of independent networks
- Held together by IP + BGP routing
- No single owner or controller

2 Mobile & cellular

- 4G LTE and 5G / 5G-Advanced
- Mobility, handover, spectrum
- The dominant path online for many

3 Data-center & cloud

- Massive internal fabrics
- Powers cloud, CDNs, AI workloads
- New topologies & congestion control

Same principles, very different scales and constraints — from a global mesh to a single warehouse of machines.

Who writes the rules of the network?

Interoperability depends on shared standards. Different bodies own different layers:



IETF

Internet protocols via RFCs — IP, TCP, QUIC, DNS



IEEE 802

LAN standards — Ethernet (802.3), Wi-Fi (802.11)



W3C / WHATWG

Web standards — HTML, HTTP semantics



3GPP

Cellular standards — 4G LTE, 5G



ITU

Global telecom coordination & spectrum



ISO

The OSI reference model, among many others

Units: reading bandwidth & data sizes

Prefix	Rate (base 10)	Size (base 2)
kilo (K)	$10^3 = 1,000$	$2^{10} = 1,024$
mega (M)	10^6	2^{20}
giga (G)	10^9	2^{30}
tera (T)	10^{12}	2^{40}

Conventions to remember

- Bandwidth uses powers of 10 (1 Mbps = 10^6 bps)
- Storage often uses powers of 2 (1 KiB = 1,024 B)
- bit (b) vs. byte (B): 1 byte = 8 bits
- Links are rated in bits/sec; files in bytes
- Watch the 8× gap: 100 Mbps $\approx$ 12.5 MB/s

Mixing bits with bytes, or base-10 with base-2, is the single most common unit mistake in networking. Read the case of the letter.

WHERE WE'RE HEADED

From one packet to the whole Internet

1

Uses & hardware — what networks are for, and the gear that builds them

2

Software & layers — protocols, layering, and the service each layer offers

3

Reference models — OSI, TCP/IP, and the 5-layer model we'll use

4

The modern Internet — QUIC, IPv6, CDNs, cloud, 5G — the 2026 reality



Next lecture: the physical layer — how bits actually cross a wire, fiber, or the air.